

Whitmore Industrials

Three analysts, three betas, three answers, and a board meeting on Tuesday.

Whitmore Industrials was a diversified manufacturer with four segments, of which one — Specialty Chemicals — was on the block. Two private equity firms and a strategic buyer had signed non-disclosure agreements. Whitmore's board wanted a view on what the segment was worth before it saw an offer, and the discount rate was the first fight.

The corporate development team had produced a beta for Whitmore of 1.15, from a regression of five years of monthly returns against the S&P 500. That was the number in the standing model, and it was the number the CFO had used in every capital budgeting memo for three years.

The banker's pitch book had a beta of 1.44, from two years of weekly returns against the same index.

An associate at one of the private equity firms, in a question sent through the data room, had cited 0.82, from five years of monthly returns against a global developed-market index rather than the S&P 500.

None of the three was computed incorrectly. Whitmore's stock had risen sharply in the last eighteen months on an unrelated litigation settlement, which was why the two-year weekly estimate ran high, and the company had a large European revenue base, which was why the global-index estimate ran low.

Amara Vieira, the corporate development director, thought all three were beside the point. Whitmore's beta described Whitmore: a conglomerate with an aerospace fastener business, a flow control business, a test-systems business, and Specialty Chemicals. The buyer was purchasing one of those four and would finance it with its own capital structure. The relevant beta was the chemicals business's, and no regression of Whitmore's stock price would ever produce it.

She pulled four pure-play specialty chemicals companies and their reported levered betas and capital structures. The buyers were both indicating a target debt-to-equity ratio near 0.45 for the standalone entity, well above Whitmore's consolidated 0.19.

ESTIMATION CHOICE	BETA	STANDARD ERROR
5 years monthly vs S&P 500	1.15	0.19
2 years weekly vs S&P 500	1.44	0.22
5 years monthly vs global developed index	0.82	0.17

EXHIBIT 2 — SPECIALTY CHEMICALS PURE PLAYS

COMPANY	LEVERED BETA	DEBT/EQUITY	MARGINAL TAX RATE
Arden Specialty	1.32	0.55	24%
Belhaven Chemical	1.05	0.28	24%
Corliss Materials	1.61	0.92	24%
Dunmore Coatings	0.94	0.19	24%

Risk-free rate: 4.2 percent. Equity risk premium: 5.0 percent. Target debt-to-equity for the standalone chemicals business: 0.45.

ASSIGNMENT

- 1 The three estimates in Exhibit 1 differ by 62 basis points of beta, which is more than 300 basis points of cost of equity. For each, say what estimation choice produced the difference and which direction it pushes.
- 2 Given the standard errors, are the three estimates statistically distinguishable from one another? Does your answer change how you would present them to the board?
- 3 Unlever the four pure plays, take a central estimate, and relever at the target capital structure. Report the cost of equity for the standalone chemicals business.
- 4 You used the median rather than the mean, or the mean rather than the median. Justify the choice. Then say what you would do if one of the four comparables had a levered beta of 2.4.
- 5 Vieira argues that Whitmore's own beta is irrelevant to this valuation. Is she right? Name the circumstance in which the company beta would be the right input.